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# AI Disclosure on Perceived Credibility in Career Advice

## Background

AI-powered career platforms are proliferating rapidly, offering students and professionals access to resume coaching, job matching, and networking advice at scale. Despite this growth, prior research suggests that people consistently discount AI-labeled advice in career contexts, preferring guidance attributed to human professionals (Chacon and Larrain, 2025). This tension raises a practical question for platform designers: **does disclosing AI as the source of career advice hurt how users perceive it?**

This study examines whether AI source attribution would negatively affect perceived credibility, advice quality, and behavioral intent compared to peer professional attribution among graduate students.

## Experimental Groups & Our Focus (gathered information)

We used a between-subjects A/B test. Participants were randomly assigned to one of two conditions via Qualtrics. Both groups read identical career advice content on transferable skills sourced from MasterCard Careers (2026). Only the attribution label differed.

The *control group* saw "Written by a MasterCard Careers professional." The *treatment group* saw the label "Generated by Claude". This design isolates the effect of AI disclosure, holding content constant.

## Hypothesis

We tested three directional hypotheses:

- H1 (Credibility): AI-attributed advice will receive significantly lower credibility and expertise ratings than peer-professional attribution.
- H2 (Quality): AI-attributed advice will be rated lower on quality and practical relevance.
- H3 (Behavioral Intent): Participants in the AI group will show lower willingness to share.

## Methods

To conduct this study, we administered a randomized controlled trial via Qualtrics. Participants were randomly assigned to either a control group, which received career advice attributed to a MasterCard Careers professional, or a treatment group, which received the same advice attributed to Claude or their self-reported favorite LLM tool. The advice content itself was identical across both conditions, only the stated source differed.

The real sample consisted of 85 Heinz graduate student respondents (41 in the AI group, 44 in the human group). To expand statistical power, 3,000 additional synthetic responses were generated by an LLM prompted to produce rows that statistically resemble the original dataset without copying existing entries, bringing the total combined dataset to 3,085 observations.

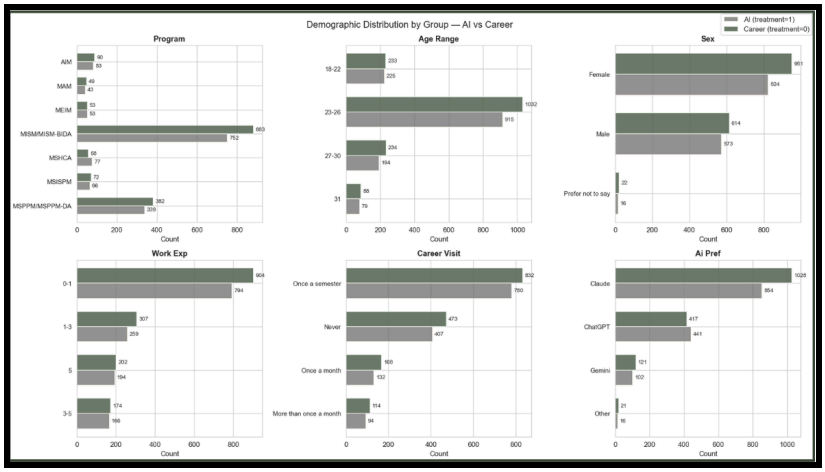



Figure 1: Demographic Distribution of the Combined Dataset (n=3,085)

Figure 1 displays the demographic distribution across both groups in the combined dataset. The sample was predominantly from the MISM/MISM-BIDA program, aged 23–26, and female. Most participants reported 0-1 years of work experience and visited the career center once a semester. Claude was the most commonly reported preferred AI tool, followed by ChatGPT, reflecting familiarity with LLM-based platforms among this graduate student population.

**Measurement Framework**

Demographic information was collected at the outset, including gender, age group, Heinz program, years of work experience, career center visit frequency, and preferred AI tool. A manipulation check asked participants to recall the source of the advice they had read, confirming treatment assignment awareness.

Four primary outcome variables were measured on 1-10 scales. Credibility was assessed with the item "I believe this advice is accurate and reliable," while expertise was captured by "The source of this advice appears to be knowledgeable about career development." Quality was measured by asking "How relevant and practical is this advice for your career?" and willingness to share by "How likely are you to share this with a friend?"

**Real & Synthetic Response**

To assess the validity of the synthetic data, we compared score distributions across all four outcome variables between the real (n=85) and LLM-generated synthetic (n=3,000) samples, as shown in Figure 2.

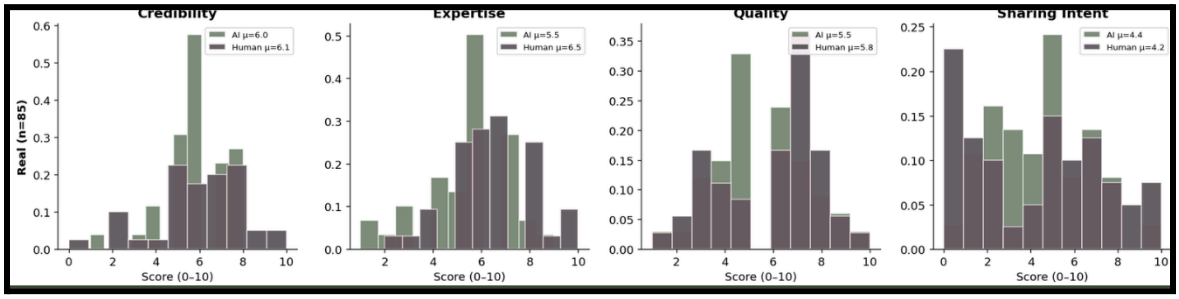



Figure 2: Score Distributions by Attribution Group of the Real Data (n=85)

In Figure 2, the real data reflects visible gaps that emerged between the AI and human attribution groups, particularly on expertise (μ=5.5 vs. μ=6.5) and quality (μ=5.5 vs. μ=5.8), with human-attributed advice consistently receiving higher ratings. Credibility showed a smaller gap (μ=6.0 vs. μ=6.1), while sharing intent was slightly higher in the AI group (μ=4.4 vs. μ=4.2).

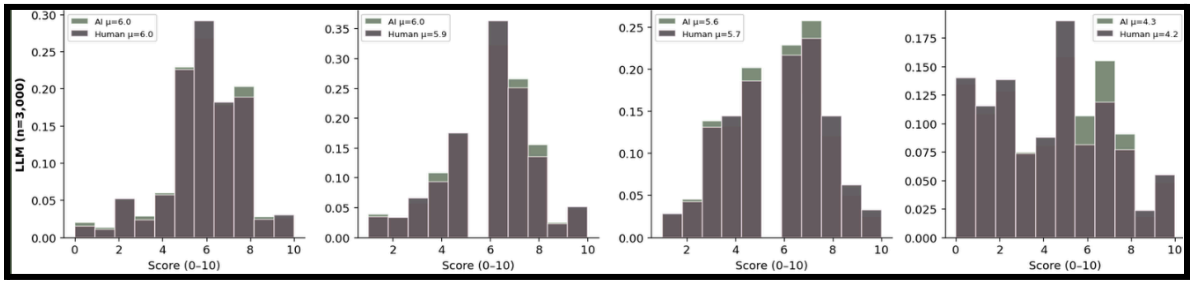



Figure 3: Score Distributions by Attribution Group of the Synthetic Data (n=3000)

Whereas in the synthetic data as shown in Figure 3, these gaps largely disappear. Mean scores across all four outcomes are nearly identical between the AI and human groups, credibility (6.0 vs. 6.0), expertise (6.0 vs. 5.9), quality (5.6 vs. 5.7), and sharing intent (4.3 vs. 4.2). This convergence suggests that the LLM was unable to replicate the attitudinal bias that real participants exhibited toward AI-attributed advice, which is an important limitation to consider when interpreting the combined dataset results.

**Summary Statistics: Human**

The real sample of 85 Heinz graduate students was evenly split between the treatment group (n=41) and the control group (n=44). Across the full sample, mean scores were 6.02 for credibility, 5.98 for expertise, 5.67 for quality, and 4.28 for sharing intent as shown in Figure 4. When broken down by group, the human-attributed condition consistently produced higher mean scores, most notably on expertise (6.45 vs. 5.46) and credibility (6.09 vs. 5.95), while sharing intent was comparable across both groups.

FULL SAMPLE SUMMARY STATISTICS

| | credibility | expertise | quality | sharing |
|---|---|---|---|---|
| count | 85.00 | 85.00 | 85.00 | 85.00 |
| mean | 6.02 | 5.98 | 5.67 | 4.28 |
| std | 1.90 | 1.85 | 2.04 | 2.89 |
| min | 0.00 | 1.00 | 1.00 | 0.00 |
| 25% | 5.00 | 5.00 | 4.00 | 2.00 |
| 50% | 6.00 | 6.00 | 6.00 | 5.00 |
| 75% | 7.00 | 7.00 | 7.00 | 7.00 |
| max | 10.00 | 10.00 | 10.00 | 10.00 |

Figure 4: Full Sample Summary Statistics of the Real Data (n=85)

| | credibility | expertise | quality | sharing |
|---|---|---|---|---|
| count | 44.00 | 44.00 | 44.00 | 44.00 |
| mean | 6.09 | 6.45 | 5.84 | 4.20 |
| std | 2.24 | 1.77 | 2.16 | 3.33 |
| min | 0.00 | 2.00 | 1.00 | 0.00 |
| 25% | 5.00 | 5.00 | 4.00 | 1.00 |
| 50% | 6.00 | 6.50 | 6.50 | 5.00 |
| 75% | 8.00 | 8.00 | 7.00 | 7.00 |
| max | 10.00 | 10.00 | 10.00 | 10.00 |

| | credibility | expertise | quality | sharing |
|---|---|---|---|---|
| count | 41.00 | 41.00 | 41.00 | 41.00 |
| mean | 5.95 | 5.46 | 5.49 | 4.37 |
| std | 1.48 | 1.80 | 1.93 | 2.37 |
| min | 1.00 | 1.00 | 1.00 | 0.00 |
| 25% | 5.00 | 4.00 | 4.00 | 2.00 |
| 50% | 6.00 | 6.00 | 5.00 | 5.00 |
| 75% | 7.00 | 7.00 | 7.00 | 6.00 |
| max | 8.00 | 9.00 | 10.00 | 10.00 |

Figure 5: Summary Statistics by Group: Control Group (n=44) and Treatment Group (n=41)

Figure 5 shows the descriptive statistics disaggregated by experimental condition. Most importantly, the human-attributed group consistently produced higher mean scores across all outcomes, most notably on expertise (6.45 vs. 5.46) and quality (5.84 vs. 5.49), while sharing intent remained comparable between groups (4.20 vs. 4.37). In addition, the AI group also showed a narrower score range, with a maximum credibility score of 8.00 compared to 10.00 in the human group.

**Summary Statistics: Synthetic**

The synthetic dataset of 3,000 responses was generated by an LLM to simulate how graduate students might rate identical career advice under AI versus human attribution conditions as shown in Figure 6. The treatment group comprised 1,413 observations and the control group 1,587 as shown in Figure 7. Mean scores across all four outcomes were nearly identical between groups, credibility (5.95 vs. 5.99), expertise (5.96 vs. 5.95), quality (5.63 vs. 5.73), and sharing intent (4.35 vs. 4.19), confirming that the synthetic data lacked the directional bias present in the real responses.

| | Credibility | Expertise | Quality | Sharing Intent |
|---|---|---|---|---|
| count | 3000.00 | 3000.00 | 3000.00 | 3000.00 |
| mean | 5.97 | 5.95 | 5.68 | 4.26 |
| std | 1.92 | 1.90 | 2.01 | 2.89 |
| min | 0.00 | 1.00 | 1.00 | 0.00 |
| 25% | 5.00 | 5.00 | 4.00 | 2.00 |
| 50% | 6.00 | 6.00 | 6.00 | 5.00 |
| 75% | 7.00 | 7.00 | 7.00 | 7.00 |
| max | 10.00 | 10.00 | 10.00 | 10.00 |

Figure 6: Full Sample Summary Statistics of the Synthetic Data (n=3000)

| | Credibility | Expertise | Quality | Sharing Intent |
|---|---|---|---|---|
| mean | 5.99 | 5.95 | 5.73 | 4.19 |
| median | 6.00 | 6.00 | 6.00 | 4.00 |
| std | 1.90 | 1.88 | 2.05 | 2.90 |
| min | 0.00 | 1.00 | 1.00 | 0.00 |
| max | 10.00 | 10.00 | 10.00 | 10.00 |

| | Credibility | Expertise | Quality | Sharing Intent |
|---|---|---|---|---|
| mean | 5.95 | 5.96 | 5.63 | 4.35 |
| median | 6.00 | 6.00 | 6.00 | 5.00 |
| std | 1.96 | 1.91 | 1.98 | 2.89 |
| min | 0.00 | 1.00 | 1.00 | 0.00 |
| max | 10.00 | 10.00 | 10.00 | 10.00 |

Figure 7: Summary Statistics by Group: Control Group (n=1587) and Treatment (n=1413)

**Balance Check**

To verify that randomization was successful, chi-square tests were conducted across five covariates, age range, gender, program, work experience, and career center usage, for both the real and synthetic datasets. All covariates were balanced in both samples, with p-values exceeding 0.05 in every case as shown in Figure 8. This confirms that any observed differences in outcomes between the AI and human attribution groups reflect the effect of the treatment itself, rather than pre-existing differences in participant characteristics.

REAL RESPONDENTS

| Covariate | Chi2 | df | p-value | Balanced? |
|---|---|---|---|---|
| Age Range | 4.119 | 3 | 0.249 | Yes |
| Gender | 0.945 | 2 | 0.623 | Yes |
| Program | 8.100 | 6 | 0.231 | Yes |
| Work Exp | 5.513 | 3 | 0.138 | Yes |
| Career Center | 2.985 | 3 | 0.394 | Yes |

LLM SYNTHETIC

| Covariate | Chi2 | df | p-value | Balanced? |
|---|---|---|---|---|
| Age Range | 1.306 | 3 | 0.728 | Yes |
| Gender | 1.363 | 2 | 0.506 | Yes |
| Program | 6.600 | 6 | 0.359 | Yes |
| Work Exp | 1.459 | 3 | 0.692 | Yes |
| Career Center | 2.788 | 3 | 0.425 | Yes |

Figure 8: Balance Check for Real and Synthetic Data

**Outcome**

Four outcomes were measured on a 0–10 scale: perceived Accuracy, Knowledgeable (perceived expertise), Relevance/Practicality, and Share Intent (willingness to share the advice).

Across the real data sample (N = 85), the AI-attributed group and the career-attributed group performed comparably on three of the four outcomes. Mean scores for Accuracy (AI: vs. Career), Relevance/Practicality, and Share Intent showed no statistically significant differences (all $p > 0.40$), suggesting that when graduate students evaluate career advice, AI attribution does not meaningfully harm their perception of its accuracy, usefulness, or their desire to share it.

The one notable exception was the **Knowledgeable** dimension. Participants rated AI-attributed advice approximately one point lower on perceived expertise than career-attributed advice (AI mean vs. Career mean: ATE = −0.99, p = 0.012). This finding indicates that while students may find AI-generated career advice just as accurate and practically useful, they are less likely to perceive the source as knowledgeable.

**Average Treatment Effects (ATE)**

| Outcome | ATE | p-value |
|---|---|---|
| Accuracy | -0.14 | > 0.40 |
| **Knowledgeable** | **−0.99** | **0.012 *** |
| Relevance/Practical | -0.35 | > 0.40 |
| Share Intent | 0.16 | > 0.40 |

The ATE for Knowledgeable (−0.99) was the only statistically significant result. The negative direction indicates that AI attribution reduced perceived expertise relative to the career-expert attribution condition. All other ATEs were small in magnitude, not statistically significant, and had 95% confidence intervals that crossed zero.

An important caveat is that synthetic data was incorporated into this analysis to supplement the real sample. However, the synthetic data failed to reproduce the group-level knowledge effect observed in the real data, indicating the synthetic augmentation matched marginal distributions but not the treatment effect itself. All primary inferences should therefore be drawn exclusively from the real data (N = 85).

**Heterogeneous Treatment Effects (HTE)**

Subgroup analyses were conducted to examine whether the treatment effect varied across demographic groups in the real dataset (N = 85, 41 treatment, 44 control).

Among gender subgroups, the negative effect on perceived knowledgeability was concentrated among female participants (ATE = −1.22, p = 0.035), while no statistically detectable effect emerged among male participants, suggesting that female graduate students may be more sensitive to AI attribution when evaluating perceived expertise as shown in Figure 9.

| | sex | Outcome | nAI | nCareer | AI mean | Career mean | ATE | 95% CI | p-value | Sig |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Female | Accuracy | 25 | 26 | 5.880000 | 5.880000 | -0.005000 | [-1.18,+1.17] | 0.993900 | n.s. |
| 1 | Female | Knowledgeable | 25 | 26 | 5.200000 | 6.420000 | -1.223000 | [-2.33,-0.12] | 0.034900 | * |
| 2 | Female | Relevance/Practical | 25 | 26 | 5.440000 | 5.690000 | -0.252000 | [-1.49,+0.99] | 0.691600 | n.s. |
| 3 | Female | Share Intent | 25 | 26 | 4.360000 | 3.810000 | 0.552000 | [-1.12,+2.22] | 0.521000 | n.s. |
| 4 | Male | Accuracy | 16 | 17 | 6.060000 | 6.470000 | -0.408000 | [-1.43,+0.61] | 0.445200 | n.s. |
| 5 | Male | Knowledgeable | 16 | 17 | 5.880000 | 6.590000 | -0.713000 | [-1.66,+0.23] | 0.150700 | n.s. |
| 6 | Male | Relevance/Practical | 16 | 17 | 5.560000 | 6.180000 | -0.614000 | [-1.78,+0.56] | 0.315200 | n.s. |
| 7 | Male | Share Intent | 16 | 17 | 4.380000 | 4.530000 | -0.154000 | [-1.93,+1.62] | 0.868300 | n.s. |

Figure 9: Heterogeneous Treatment Effect on Real Data (Gender)

By age, the 23–26 cohort showed the strongest negative treatment effect on Knowledgeability (ATE = −1.37, p = 0.006), while no other age stratum reached significance as shown in Figure 10. No significant

heterogeneous treatment effects were observed for Accuracy, Relevance/Practicality, or Share Intent across either gender or age subgroups. Given the modest sample sizes within individual strata, these findings should be treated as exploratory rather than confirmatory. No significant heterogeneous treatment effects were observed for Accuracy, Relevance/Practicality, or Share Intent across either gender or age subgroups.

| | age_range | Outcome | nAI | nCareer | AI mean | Career mean | ATE | 95% CI | p-value | Sig |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 18-22 | Accuracy | 4 | 9 | 5.000000 | 5.780000 | -0.778000 | [-2.89,+1.33] | 0.626600 | n.s. |
| 1 | 18-22 | Knowledgeable | 4 | 9 | 5.750000 | 7.000000 | -1.250000 | [-3.05,+0.55] | 0.279300 | n.s. |
| 2 | 18-22 | Relevance/Practical | 4 | 9 | 5.250000 | 5.670000 | -0.417000 | [-2.03,+1.19] | 0.737500 | n.s. |
| 3 | 18-22 | Share Intent | 4 | 9 | 5.000000 | 3.440000 | 1.556000 | [-1.35,+4.46] | 0.380300 | n.s. |
| 4 | 23-26 | Accuracy | 29 | 26 | 6.000000 | 6.230000 | -0.231000 | [-1.23,+0.76] | 0.647200 | n.s. |
| 5 | 23-26 | Knowledgeable | 29 | 26 | 5.240000 | 6.620000 | -1.374000 | [-2.31,-0.44] | 0.006300 | ** |
| 6 | 23-26 | Relevance/Practical | 29 | 26 | 5.480000 | 6.230000 | -0.748000 | [-1.84,+0.35] | 0.187300 | n.s. |
| 7 | 23-26 | Share Intent | 29 | 26 | 4.070000 | 4.920000 | -0.854000 | [-2.44,+0.73] | 0.288800 | n.s. |
| 8 | 27-30 | Accuracy | 7 | 5 | 6.290000 | 6.200000 | 0.086000 | [-2.37,+2.54] | 0.938500 | n.s. |
| 9 | 27-30 | Knowledgeable | 7 | 5 | 6.000000 | 5.600000 | 0.400000 | [-1.95,+2.75] | 0.730600 | n.s. |
| 10 | 27-30 | Relevance/Practical | 7 | 5 | 5.710000 | 4.800000 | 0.914000 | [-1.72,+3.55] | 0.484400 | n.s. |
| 11 | 27-30 | Share Intent | 7 | 5 | 5.140000 | 1.400000 | 3.743000 | [+1.78,+5.70] | 0.006100 | ** |

Figure 10: Heterogeneous Treatment Effect on Real Data (Age)

To assess robustness, the same analysis was repeated on the full dataset combining real and synthetically augmented observations (N = 3,085; 1,454 treatment, 1,631 control).

By gender, among female participants, the AI condition produced a statistically significant negative effect on perceived Accuracy (ATE = −0.217, p = 0.018), while effects on Knowledgeable, Relevance/Practicality, and Share Intent remained non-significant as shown in Figure 12. Among male participants, the AI condition yielded a significant positive effect on Accuracy (ATE = +0.254, p = 0.018) and a significant negative effect on Relevance/Practicality (ATE = −0.251, p = 0.026); all other outcomes were non-significant.

By age, the 23–26 age group again showed the only statistically significant subgroup effect: a negative impact on Relevance/Practicality (ATE = −0.245, p = 0.007). No significant effects were detected in any other age group (18–22, 27–30, or 31+) across any outcome as shown in Figure 11.

| | age_range | Outcome | nAI | nCareer | AI mean | Career mean | ATE | 95% CI | p-value | Sig |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 18-22 | Accuracy | 229 | 242 | 5.970000 | 6.090000 | -0.126000 | [-0.48,+0.22] | 0.480800 | n.s. |
| 1 | 18-22 | Knowledgeable | 229 | 242 | 5.850000 | 6.080000 | -0.235000 | [-0.58,+0.11] | 0.183900 | n.s. |
| 2 | 18-22 | Relevance/Practical | 229 | 242 | 5.720000 | 5.580000 | 0.138000 | [-0.23,+0.51] | 0.467800 | n.s. |
| 3 | 18-22 | Share Intent | 229 | 242 | 4.330000 | 4.340000 | -0.011000 | [-0.52,+0.50] | 0.965300 | n.s. |
| 4 | 23-26 | Accuracy | 944 | 1058 | 5.980000 | 5.980000 | -0.001000 | [-0.17,+0.17] | 0.994300 | n.s. |
| 5 | 23-26 | Knowledgeable | 944 | 1058 | 5.930000 | 5.980000 | -0.044000 | [-0.21,+0.12] | 0.604400 | n.s. |
| 6 | 23-26 | Relevance/Practical | 944 | 1058 | 5.580000 | 5.820000 | -0.245000 | [-0.42,-0.07] | 0.006900 | ** |
| 7 | 23-26 | Share Intent | 944 | 1058 | 4.300000 | 4.170000 | 0.138000 | [-0.12,+0.39] | 0.286600 | n.s. |
| 8 | 27-30 | Accuracy | 201 | 239 | 5.890000 | 5.900000 | -0.010000 | [-0.36,+0.34] | 0.956200 | n.s. |
| 9 | 27-30 | Knowledgeable | 201 | 239 | 6.060000 | 5.770000 | 0.295000 | [-0.06,+0.65] | 0.096600 | n.s. |
| 10 | 27-30 | Relevance/Practical | 201 | 239 | 5.790000 | 5.590000 | 0.197000 | [-0.16,+0.56] | 0.287500 | n.s. |
| 11 | 27-30 | Share Intent | 201 | 239 | 4.370000 | 4.110000 | 0.259000 | [-0.30,+0.82] | 0.367800 | n.s. |
| 12 | 31 | Accuracy | 80 | 92 | 5.790000 | 6.110000 | -0.321000 | [-0.94,+0.30] | 0.307500 | n.s. |
| 13 | 31 | Knowledgeable | 80 | 92 | 6.010000 | 5.900000 | 0.110000 | [-0.47,+0.69] | 0.710200 | n.s. |
| 14 | 31 | Relevance/Practical | 80 | 92 | 5.500000 | 5.390000 | 0.109000 | [-0.48,+0.69] | 0.721800 | n.s. |
| 15 | 31 | Share Intent | 80 | 92 | 4.860000 | 4.240000 | 0.623000 | [-0.23,+1.48] | 0.156900 | n.s. |

Figure 11:  Heterogeneous Treatment Effect on Combined Data (Age)

| | sex | Outcome | nAI | nCareer | AI mean | Career mean | ATE | 95% CI | p-value | Sig |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Female | Accuracy | 849 | 977 | 5.860000 | 6.070000 | -0.217000 | [-0.40,-0.04] | 0.017900 | * |
| 1 | Female | Knowledgeable | 849 | 977 | 5.900000 | 5.940000 | -0.043000 | [-0.22,+0.13] | 0.624300 | n.s. |
| 2 | Female | Relevance/Practical | 849 | 977 | 5.650000 | 5.670000 | -0.016000 | [-0.20,+0.17] | 0.866200 | n.s. |
| 3 | Female | Share Intent | 849 | 977 | 4.340000 | 4.150000 | 0.188000 | [-0.08,+0.45] | 0.166000 | n.s. |
| 4 | Male | Accuracy | 589 | 631 | 6.110000 | 5.850000 | 0.254000 | [+0.04,+0.46] | 0.018000 | * |
| 5 | Male | Knowledgeable | 589 | 631 | 6.000000 | 5.990000 | 0.011000 | [-0.20,+0.23] | 0.919800 | n.s. |
| 6 | Male | Relevance/Practical | 589 | 631 | 5.580000 | 5.830000 | -0.251000 | [-0.47,-0.03] | 0.026000 | * |
| 7 | Male | Share Intent | 589 | 631 | 4.350000 | 4.200000 | 0.150000 | [-0.17,+0.47] | 0.362500 | n.s. |

Figure 12: Heterogeneous Treatment Effect on Combined Data (Gender)

**Power Calculation and Limitations**

**1. Underpowered Sample for Most Outcomes** The most significant limitation of this study is statistical power. As shown in Figure 13, with approximately 41–44 participants per group, only the Knowledgeable outcome achieved near-adequate power (power ≈ 0.714). The remaining three outcomes (Accuracy, Relevance/Practicality, and Share Intent) were severely underpowered (observed power < 0.15 for each). Power analyses indicate that between 53 and 5,000+ participants per group would be required to reliably detect effects of the magnitudes observed, depending on the outcome. As a result, null findings for those three outcomes should not be interpreted as evidence of no effect.

| | Outcome | Cohen's d | Effect | Observed power | 80% powered? | N/group for 80% | N/group for 90% | Current n/group |
|---|---|---|---|---|---|---|---|---|
| 0 | Accuracy | 0.073000 | small | 0.063000 | No ✗ | 2943 | 3940 | 41 |
| 1 | Knowledgeable | 0.555000 | large | 0.714000 | No ✗ | 53 | 70 | 41 |
| 2 | Relevance/Practical | 0.172000 | small | 0.123000 | No ✗ | 530 | 708 | 41 |
| 3 | Share Intent | 0.055000 | small | 0.057000 | No ✗ | 5109 | 6839 | 41 |

Figure 13: Power Calculation

**2. Small Subgroup Sizes in HTE Analyses** Heterogeneous treatment effect analyses by gender and age group were conducted on subsets of an already small sample. Several subgroup cells contained fewer than 10 participants per condition, which makes subgroup ATEs highly unstable and the p-values unreliable. The HTE findings are best treated as hypothesis-generating rather than definitive.

**3. Synthetic Data Does Not Reproduce Treatment Effects** The synthetic data augmentation, while useful for exploring distributional properties, diluted the observed treatment effect on Knowledgeable from a Cohen's d of 0.555 (real data) to approximately 0.008 (synthetic data). This discrepancy indicates the synthetic data generation process did not adequately model the treatment-response relationship. Any analyses using the combined or synthetic-only datasets should be interpreted with extreme caution and should not substitute for real-data findings.

**4. Minor Covariate Imbalance** Randomization produced well-balanced groups on most demographic variables. However, work experience showed a modest imbalance between conditions (maximum proportion gap = 0.173). While this variable is not strongly theorized to confound AI credibility perception, it introduces some uncertainty around the clean causal interpretation of the estimates.

**5. Sample and Generalizability** The study was conducted on graduate students, which limits generalizability to other populations. Graduate students may have higher baseline familiarity with AI tools and different trust calibration relative to the general public or other professional groups.

## Conclusion

AI disclosure negatively impacts perceived expertise but does not significantly affect perceived accuracy, relevance, or willingness to share career advice. Specifically, career advice attributed to AI is rated approximately one point lower on perceived knowledgeability compared to the same advice attributed to a human career expert. This effect appears to be stronger among female participants and those in the 23–26 age range. However, these results are preliminary: the sample is underpowered for most outcomes, subgroup findings are exploratory, and the synthetic data augmentation was not reliable for causal inference. Replication with a larger, adequately powered sample is needed before drawing firm conclusions.

## References

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